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STAT-341 Final Project Presentation

4/25/23

Eye Contact Marks The Rise And Fall of Shared Attention in Conversation

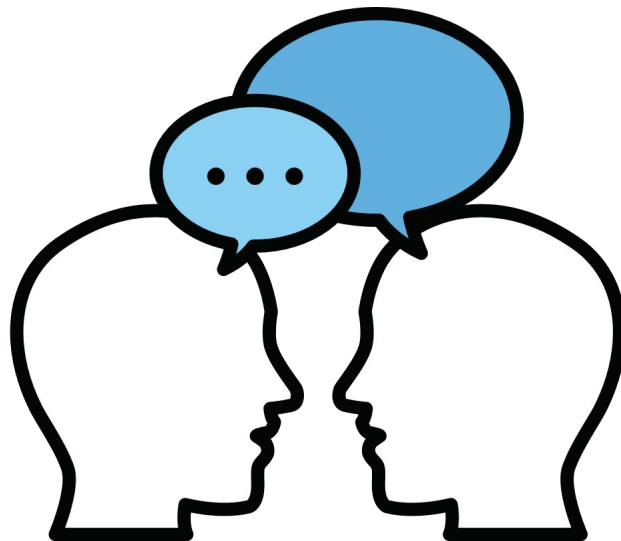
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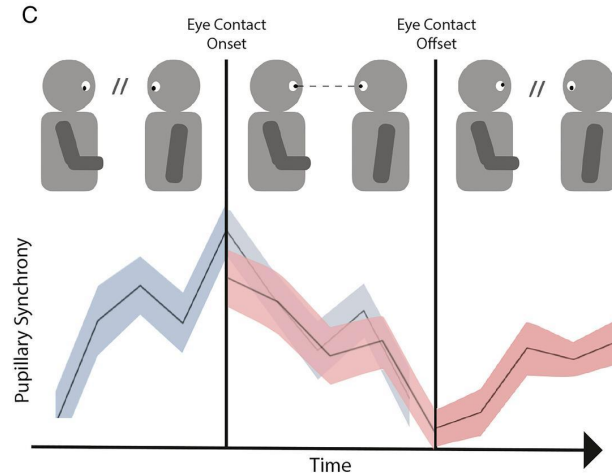
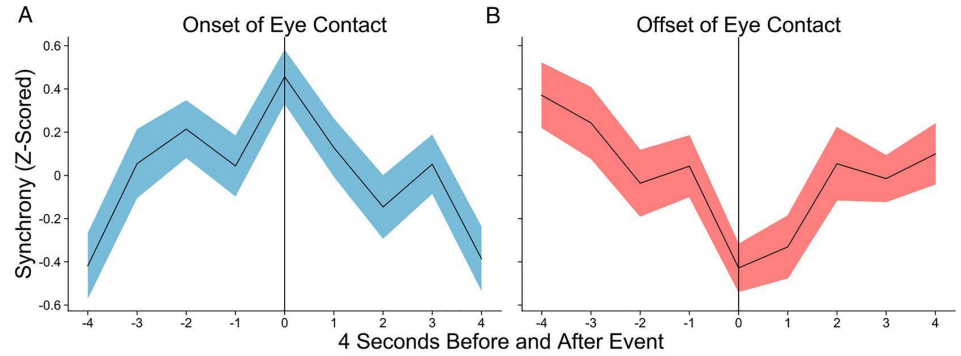
Research Background

The paper explores how two minds coordinate during a conversation, with a focus on the role of eye contact.

When two people converse, their pupils periodically **synchronize**, marking moments of **shared attention**.



As synchrony peaks, eye contact occurs and synchrony declines, only to recover as eye contact breaks.



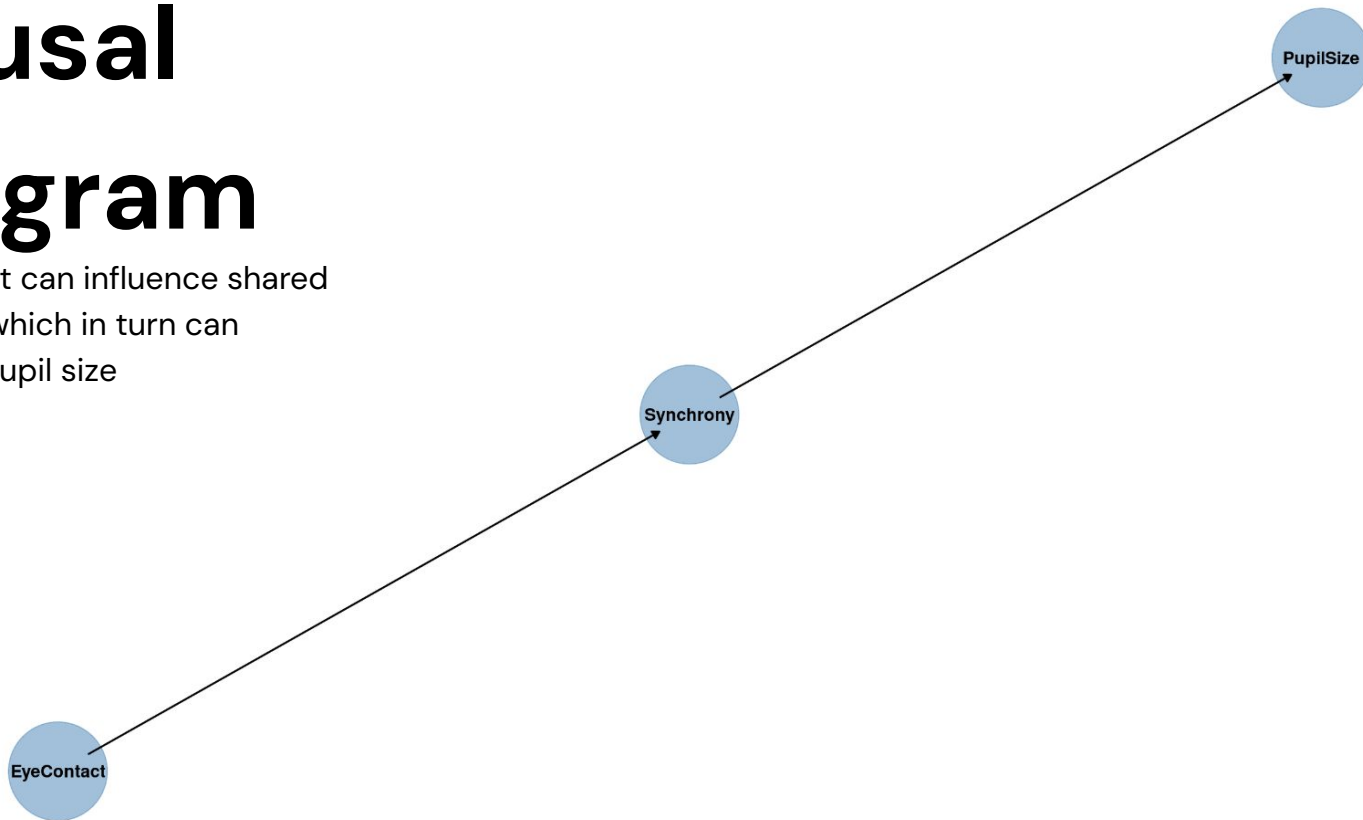
Dataset

	X	dyad	subject	dtw	EyeContact	PupilSize	Synchrony	Sync_standardized
1	0	D001	S028	26.038945	0.000	3.509223e-01	-3.2595933	-0.511344522
2	1	D001	S028	42.312825	0.000	4.793972e-01	-3.7450902	-1.084150890
3	2	D001	S028	27.318998	0.033	4.749346e-01	-3.3075824	-0.567963705
4	3	D001	S028	37.486558	0.256	5.588322e-01	-3.6239824	-0.941263615
5	4	D001	S028	15.107104	0.350	3.743855e-01	-2.7151651	0.130991095
6	5	D001	S028	21.026297	0.000	4.666214e-01	-3.0457739	-0.259072831
7	6	D001	S028	10.646218	0.000	4.121449e-01	-2.3652047	0.543886695
8	7	D001	S028	31.755402	0.085	2.421469e-01	-3.4580628	-0.745505873
9	8	D001	S028	2.090744	1.000	2.109000e-01	-0.7375202	2.464286222
10	9	D001	S028	4.118150	0.566	8.580097e-02	-1.4154039	1.664495085
11	10	D001	S028	5.505230	0.000	1.750619e-01	-1.7056985	1.321995266
12	11	D001	S028	8.040730	0.765	2.097454e-02	-2.0845199	0.875048498
13	12	D001	S028	5.922619	0.896	7.121833e-02	-1.7787787	1.235772684
14	13	D001	S028	10.958424	0.746	2.367272e-01	-2.3941085	0.509784991
15	14	D001	S028	4.880071	0.714	6.403745e-02	-1.5851597	1.464211201
16	15	D001	S028	9.280057	0.000	1.400984e-01	-2.2278677	0.705921748
17	16	D001	S028	12.503160	0.000	1.622589e-01	-2.5259814	0.354196632
18	17	D001	S028	9.631593	0.000	6.068541e-02	-2.2650486	0.662054367
19	18	D001	S028	14.923801	0.000	9.592979e-02	-2.7029573	0.145394196
20	19	D001	S028	20.001669	0.936	3.707470e-01	-2.9958157	-0.200130438

Showing 1 to 20 of 56,400 entries, 12 total columns

Causal Diagram

Eye contact can influence shared attention, which in turn can influence pupil size



Our Model

$$\text{Sync_standardized}_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 * \text{EyeContact}_i + \beta_2 * \text{PupilSize}_i$$

$$\beta_0 \sim \text{Normal}(0, 0.5)$$

$$\beta_1 \sim \text{Normal}(0.1, 0.5)$$

$$\beta_2 \sim \text{Normal}(0, 1)$$

$$\sigma \sim \text{lognormal}(0, 1)$$

STAN

Results

```
mlvar_stan_model <- stan(model_code = stan_mlvar_program,  
                          data = stan_data,  
                          chains = 4,  
                          iter = 1500,  
                          warmup = 750,  
                          refresh = 0)
```

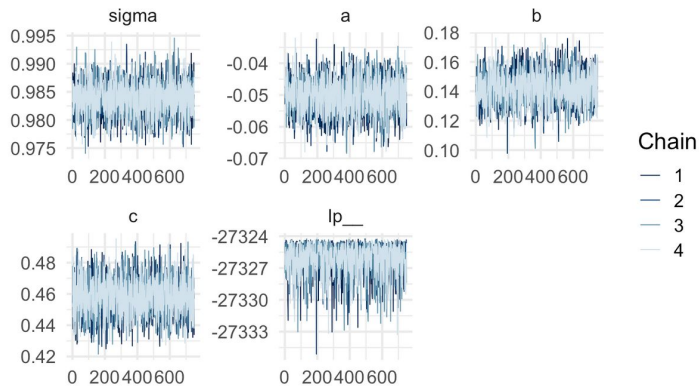
```
mlvar_stan_model
```

```
## Inference for Stan model: 56de5ddbf84ffa35e6d9c01fc12bae1b.  
## 4 chains, each with iter=1500; warmup=750; thin=1;  
## post-warmup draws per chain=750, total post-warmup draws=3000.  
##  
##           mean se_mean  sd      2.5%      25%      50%      75%      97.5%  
## sigma      0.98      0.00 0.00      0.98      0.98      0.98      0.99      0.99  
## a        -0.05      0.00 0.01     -0.06     -0.05     -0.05     -0.05     -0.04  
## b         0.14      0.00 0.01      0.12      0.14      0.14      0.15      0.16  
## c         0.46      0.00 0.01      0.44      0.45      0.46      0.47      0.48  
## lp__    -27326.21      0.04 1.42    -27329.92    -27326.90    -27325.88    -27325.15    -27324.47  
##           n_eff Rhat  
## sigma    3027      1  
## a         2025      1  
## b         1808      1  
## c         2303      1  
## lp__     1453      1  
##  
## Samples were drawn using NUTS(diag_e) at Mon Apr 24 20:39:44 2023.  
## For each parameter, n_eff is a crude measure of effective sample size,  
## and Rhat is the potential scale reduction factor on split chains (at  
## convergence, Rhat=1).
```

Convergence Diagnostics

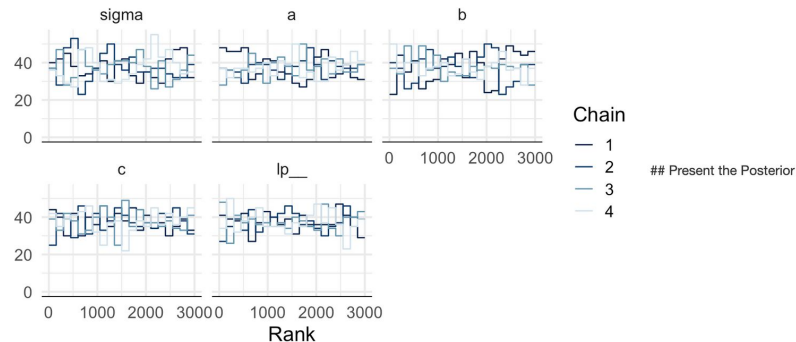
Trace Plot

```
bayesplot::mcmc_trace(mlvar_stan_model)
```



Rank Trace Plot

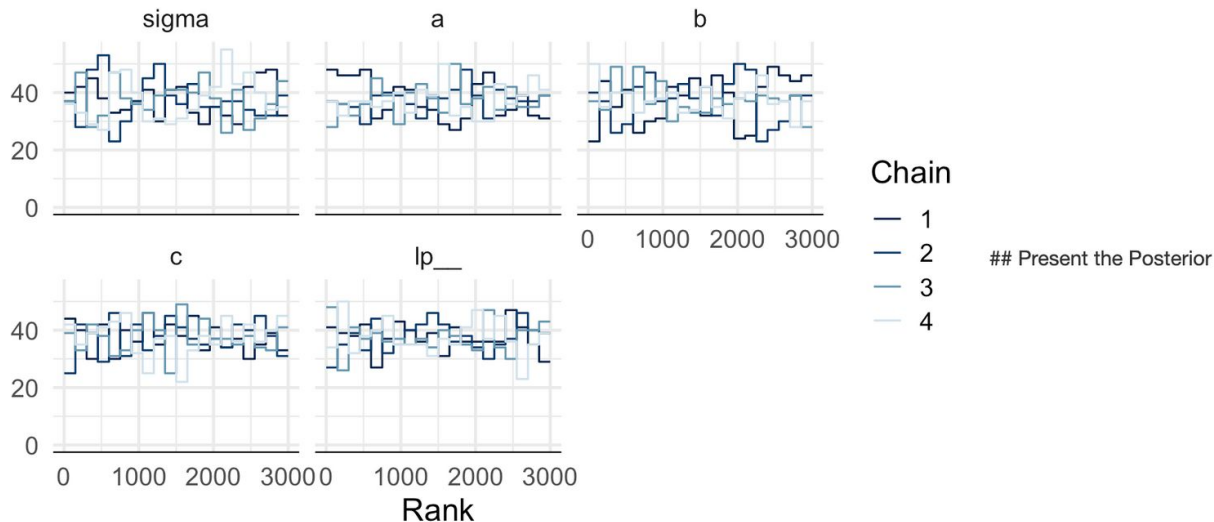
```
bayesplot::mcmc_rank_overlay(mlvar_stan_model)
```



Convergence Diagnostics

Rank Trace Plot

```
bayesplot::mcmc_rank_overlay(mlvar_stan_model)
```



Add – Ons

Categorical Variable

```
set.seed(50)
mlvar_data$attention <- rbinom(n = nrow(mlvar_data), size
= 1, prob = 0.7) + 1
## 1: not engage, 2: engage.
```

attention
2
2
1
2

Hierarchical Model

```
```{r}
personLevelData <- mlvar_data |>
 group_by(subjectNum) |>
 summarise(subjectNum = first(subjectNum))|>
 ungroup()

glimpse(personLevelData)
```
```

Rows: 94
Columns: 1
\$ subjectNum <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25,
26, 27, 28, ...

Prior Predictive Check

